Md Miadul Islam Nizzan

AI/ML & NLP | Machine Learning Developer | Undergraduate Researcher

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PROFILE

Final-year Computer Science and Engineering undergraduate at Green University of Bangladesh with applied research and project experience in machine learning, Bangla NLP, and computer vision. Skilled in Python-based data preparation, Transformer experimentation, image classification, and model evaluation. Experienced in developing AI-enabled applications and communicating technical work through research writing, documentation, mentoring, and presentations.

EDUCATION

Green University of Bangladesh — Expected January 2027
B.Sc. in Computer Science and Engineering · Rupganj, Narayanganj, Bangladesh

- Current CGPA: 3.71 / 4.00
- Dean's Award for outstanding academic performance
- Vice Chancellor's Award for academic excellence

Royal Media College — 2021
Higher Secondary Certificate (HSC) · Mymensingh, Bangladesh

Shaheen Cadet School — 2019
Secondary School Certificate (SSC) · Bhaluka, Mymensingh, Bangladesh

RESEARCH

Robustness, Explainability, and Attribution Stability of Lightweight Vision Transformers for Federated Chest X-ray Classification under Non-IID Client Distributions — GitHub

Undergraduate Thesis · Ongoing, 2026 – present

A federated binary-classification study for pleural effusion detection using NIH ChestX-ray14 across 10 simulated non-IID hospital clients.

- Comparing centralized, local-only, FedAvg, and FedProx training with DeiT-Tiny and ResNet-18
- Noisy-label robustness and attribution-stability analysis
- Grad-CAM and Vision Transformer attribution methods to evaluate explanation consistency across client distributions

Robustness and Tokenization Stability of Transformer Models for 12-Class Bangla Dialect Classification Under Character-Level Perturbations

Manuscript · Submitted — decision pending

Evaluated BanglaBERT, multilingual DistilBERT and XLM-RoBERTa on 59,453 BanglaDial records under progressive 0–20% character-level perturbations.

- Multilingual DistilBERT was most robust at the 20% perturbation condition
- Typo-aware augmentation improved its accuracy from 51.005% to 62.913%

Multilevel Detection of Social Media Harassment Targeting Women

Research in Progress · Dataset development and text-classification model fine-tuning

Curating a multilevel dataset of social-media harassment targeting women and developing a structured labeling scheme for model training and evaluation.

- Multilevel dataset curation and a structured labeling scheme
- Fine-tuning a text-classification model
- Preparing the study for a future research manuscript, subject to experimental results

TECHNICAL EXPERTISE

AI, Machine Learning & NLP	TensorFlow, Keras, PyTorch, Hugging Face Transformers, scikit-learn, LangChain, RAG pipelines, Multi-agent workflows, CNNs, MobileNetV2, BERT-based NLP, Bangla NLP, BanglaBERT, XLM-RoBERTa, Text classification, Transfer learning, Model evaluation
Agentic AI	LangChain, Groq API, Tavily search, Tool calling, Role-specific prompting, Source attribution, Streamlit deployment
Federated & Explainable AI	FedAvg, FedProx, Non-IID learning, DeiT-Tiny, ResNet-18, Grad-CAM, Vision Transformer attribution, Robustness analysis
Programming	Python, JavaScript, TypeScript, Dart, PHP, SQL, C (working knowledge), C++ (working knowledge), Java (working knowledge), x86 Assembly (working knowledge), Data structures & algorithms (fundamentals)

Web & Mobile	React, Vite, Node.js, Django REST Framework, Flask, Flutter, REST APIs, HTML/CSS, Authentication, Server-Sent Events
Data & Databases	Pandas, NumPy, OpenCV, MySQL, PostgreSQL, SQLite, Supabase, Data preprocessing, Feature engineering
Security & Networking	TCP/IP, Packet analysis, AES, RSA, SHA-256, Network fundamentals, Cryptographic verification
Tools	Git, GitHub, Jupyter, Google Colab, VS Code, NetBeans, XAMPP, LaTeX, Microsoft Office

SELECTED PROJECTS

AI Research Assistant Agent GitHub | Live demo
Python, LangChain, Groq, Tavily, Streamlit
- Web-search tool calling with role-specific prompts for each agent
- Explicit failure handling and saved reports

Smart Farming System: Bengali-First Agricultural Platform GitHub
React, TypeScript, Vite, Node.js, PostgreSQL
- Plant-disease detection, automated irrigation control, and expert consultation workflows
- Market-price verification and multi-provider AI chat

CogniSearch: Research-Oriented AI Chatbot GitHub
Python, Streamlit, Groq API, Llama 3.3
- Temperature, Top-P, and token controls with persistent conversation state
- Markdown export and secure credential configuration

Fosholer Bondhu: AI Agricultural Assistant GitHub
Python, TensorFlow, Keras, MobileNetV2, TensorFlow Lite
- MobileNetV2 transfer-learning workflow for disease classification
- Structured for evaluation and TensorFlow Lite deployment on-device

Farm Operation & Transaction Management GitHub
Flutter, Django REST Framework, JWT, SQLite, PostgreSQL
- Offline-first data model for use in low-connectivity settings
- End-to-end coverage from animal health records to financial reporting

Secure File Sharing & Cryptographic Verification GitHub
Python, Flask, AES-128-CBC, RSA, SHA-256
- Cryptographic primitives implemented from first principles, not library calls
- Role-based access control with tamper simulation and audit logging

EXPERIENCE & LEADERSHIP

Vice Chair · IEEE GUB Student Branch, Executive Committee 2026 – Present
Coordinate technical events, workshops, student initiatives, communications, and cross-team collaboration.

Student Mentor · Green University of Bangladesh 2025 – Present
Guide junior students in academic planning, technical coursework, projects, GitHub, presentations, and documentation.

Assistant Graphic Designer · IEEE GUB Student Branch, Executive Committee 2025 – 2026
Supported branding, event promotion, technical program publicity, and digital content development.

Assistant Teacher · Kolatoli Darus Sunnah Al Arabiah Madrashah 2023 – 2024
Delivered structured lessons, provided individual guidance, and managed classroom activities and student development.

AWARDS, MEMBERSHIP & SERVICE

- **Vice Chancellor's Award** · Green University of Bangladesh
- **Dean's Award** · Green University of Bangladesh
- **Debate Champion** · Royal Media College (2020)
- **IEEE Member** · IEEE (2025)
- **IEEE Day Ambassador** · IEEE GUB Student Branch
- **STI 2025 Conference Volunteer** · IEEE (2025)